

DSA End Module Exam

Total points 36/40 ?

The respondent's email (omjbhaleraowork@gmail.com) was recorded on submission of this form.

✓ The number of comparisons done by sequential search is* 1/1

- ☐ $n/2)+1$
- ☒ $(n+1)/2$
- ☐ $(n-1)/2$
- ☐ $(n+2)/2$



✗ node.next -> node.next.next; will make * 0/1

- ☐ node.next inaccessible
- ☐ node.next.next inaccessible
- ☒ this node inaccessible
- ☐ none of the above



Correct answer

- ☒ node.next inaccessible



✓ Which of the following data structure is required to convert arithmetic expression in infix to its equivalent postfix notation? *1/1

- ☐ Queue
- ☐ Linked list
- ☐ Binary search tree
- ☒ None of above



✓ Suppose we have an $O(n)$ time algorithm that finds the median of an unsorted array. Now consider a QuickSort implementation where we first find the median using the above algorithm, then use the median as a pivot. What will be the worst-case time complexity of this modified QuickSort? *1/1

- ☐ $O(n^2 \text{ Log} n)$
- ☐ $O(n^2)$
- ☐ $O(n \text{ Log} n \text{ Log} n)$
- ☒ $O(n \text{ Log} n)$



✓ Which data structure do we use for testing a palindrome? * 1/1

- ☐ Heap
- ☐ Tree
- ☐ Priority queue
- ☒ Stack



✓ Which data structure is used in breadth first search of a graph to hold nodes? *1/1

☐ stack

☒ queue

☐ tree

☐ array



✓ Suppose the numbers 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are inserted in that order into an initially empty binary search tree. The binary search tree uses the usual ordering on natural numbers. What is the in-order traversal sequence of the resultant tree? *1/1

☐ 7 5 1 0 3 2 4 6 8 9

☐ 0 2 4 3 1 6 5 9 8 7

☒ 0 1 2 3 4 5 6 7 8 9

☐ 9 8 6 4 2 3 0 1 5 7



✗ The time complexity used for inserting a node in a priority queue on the basis of key is: *0/1

☐ $O(n)$

☐ $O(n^2)$

☐ $O(n \log n)$

☒ $O(\log n)$



Correct answer

☒ $O(n)$



✓ Quick sort is also known as *

1/1

- ☐ merge sort
- ☐ tree sort
- ☐ shell sort
- ☒ partition and exchange sort



✓ In what order will they be removed If the elements "A", "B", "C" and "D" are placed in a queue and are deleted one at a time *1/1

- ☒ ABCD
- ☐ DCAB
- ☐ DCBA
- ☐ ABDC



✓ In a priority queue, insertion and deletion takes place at *

1/1

- ☐ front, rear end
- ☐ only at rear end
- ☐ only at front end
- ☒ any position



✓ A directed graph is if there is a path from each vertex to every other vertex in the digraph. *1/1

- ☐ weakly connected
- ☒ strongly connected
- ☐ tightly connected
- ☐ linearly connected



✓ For Breadth-First Traversal on a graph is the data structure required? * 1/1

- ☐ Stack
- ☒ queue
- ☐ array
- ☐ Tree
- ☐ Both a&b



✓ What is the type of the algorithm used in solving the 8 Queens problem? * 1/1

- ☒ Backtracking
- ☐ Dynamic
- ☐ Branch and Bound
- ☐ None



Last TWO Digits of PRN *

48
.....

✓ . The result of evaluating prefix expression $*/b+-dacd$, where $a = 3$, $b = 6$, $c = 1$, $d = 5$ is *1/1

- ☐ 0
- ☐ 5
- ☒ 10
- ☐ 15



✓ The time factor when determining the efficiency of algorithm is measured by *1/1

- ☐ Counting microseconds
- ☒ Counting the number of key operations
- ☐ Counting the number of statements
- ☐ Counting the kilobytes of algorithm



✓ A binary search tree whose left subtree and right subtree differ in height by ^{*}1/1 at most 1 unit is called

- ☒ avl tree
- ☐ red-black tree
- ☐ lemma tree
- ☐ none of the above



✓ Recursive algorithms are based on ^{*}

1/1

- ☐ Divide and conquer approach
- ☐ Top-down approach
- ☒ Bottom-up approach
- ☐ Hierarchical approach



✗ What is the equivalent infix expression of the following postfix expression?

^{*}0/1

M, N, O, +, *, P, /, Q, R, S, T, /, +, *, -

- ☐ $N * (M + O) / Q - P * (S + R / T)$
- ☐ $((((M * (N + O)) / P) - (Q * (R + (S / T))))$
- ☒ $O * (M + N) / P - Q * (R + S / T)$
- ☐ $M * (N + O) / Q - P * (R / S + T)$



Correct answer

- ☒ $((((M * (N + O)) / P) - (Q * (R + (S / T))))$



✓ In the traversal we process all of a vertex's descendants before we move to an adjacent vertex. *1/1

- ☒ depth first
- ☐ breadth first
- ☐ with first
- ☐ depth limited



✓ The data structure which is one ended is * 1/1

- ☐ queue
- ☒ stack
- ☐ tree
- ☐ graph



✓ A graph is a tree if and only if graph is * 1/1

- ☐ Directed graph
- ☒ Contains no cycles
- ☐ Planar
- ☐ Completely connected



✓ The complexity of Bubble sort algorithm is *

1/1

- ☐ $O(n)$
- ☐ $O(\log n)$
- ☒ $O(n^2)$
- ☐ $O(n \log n)$



The disadvantage in using a circular linked list is

*

1/1

- ☒ it is possible to get into infinite loop.
- ☐ last node points to first node.
- ☐ time consuming
- ☐ requires more memory space



✓ Dijkstra's algorithm is used to solve _____ problems? *

1/1

- ☐ Network lock
- ☒ Single source shortest path
- ☐ All pair shortest path
- ☐ Sorting



✓ A graph in which all vertices have equal degree is known as ____ * 1/1

- ☒ Complete graph
- ☐ Regular graph
- ☐ Multi graph
- ☐ Simple graph



✓ The post order traversal of a binary tree is DEBFCA. Find out the pre order *1/1 traversal

- ☐ ABFCDE
- ☐ ADBFEC
- ☒ ABDECF
- ☐ ABDCEF



Full Name *

Om Jairam Bhalerao

✓ The time complexity of quick sort is * 1/1

- ☐ $O(n)$
- ☐ $O(n^2)$
- ☒ $O(n \log n)$
- ☐ $O(\log n)$



✓ Assuming int is of 4bytes, what is the size of int arr[15];? *

1/1

- ☐ 15
- ☐ 19
- ☐ 11
- ☒ 60



✓ In a graph if $e=[u, v]$, Then u and v are called *

1/1

- ☐ endpoints of e
- ☐ adjacent nodes
- ☐ neighbors
- ☒ all of above



✓ A program P reads in 500 integers in the range [0..100] representing the scores of 500 students. It then prints the frequency of each score above 50. What would be the best way for P to store the frequencies? *1/1

- ☒ An array of 50 numbers
- ☐ An array of 100 numbers
- ☐ An array of 500 numbers
- ☐ A dynamically allocated array of 550 numbers



✓ Insertion of an element at the middle of a linked list requires the modification of how many pointers?

*1/1

☒ 2



☐ 1

☐ 3

☐ 4

✗ Suppose a circular queue of capacity $(n - 1)$ elements is implemented with an array of n elements. Assume that the insertion and deletion operations are carried out using REAR and FRONT as array index variables, respectively. Initially, $\text{REAR} = \text{FRONT} = 0$. The conditions to detect queue full and queue empty are

*0/1

☐ full: $(\text{REAR} + 1) \bmod n == \text{FRONT}$ empty: $\text{REAR} == \text{FRONT}$

☒ full: $(\text{REAR} + 1) \bmod n == \text{FRONT}$ empty: $(\text{FRONT} + 1) \bmod n == \text{REAR}$



☐ full: $\text{REAR} == \text{FRONT}$ empty: $(\text{REAR} + 1) \bmod n == \text{FRONT}$

☐ full: $(\text{FRONT} + 1) \bmod n == \text{REAR}$ empty: $\text{REAR} == \text{FRONT}$

Correct answer

☒ full: $(\text{REAR} + 1) \bmod n == \text{FRONT}$ empty: $\text{REAR} == \text{FRONT}$



✓ if implemented using an array of size MAX_SIZE, gets full when *

1/1

- ☐ Front = (rear + 1)mod MAX_SIZE
- ☐ Front = rear + 1
- ☒ Rear = MAX_SIZE - 1
- ☐ Rear = front
- ☐ None of above



✓ How can we describe an array in the best possible way? *

1/1

- ☐ The Array shows a hierarchical structure.
- ☐ Arrays are immutable.
- ☒ Container that stores the elements of similar types
- ☐ The Array is not a data structure



✓ Finding the location of the element with a given value is: *

1/1

- ☐ traversal
- ☒ search
- ☐ sort
- ☐ none of above



✓ What is the functionality of the following code? *

1/1

```
public void function(Node node)
{
    if(size == 0)
        head = node;
    else
    {
        Node temp, cur;
        for(cur = head; (temp = cur.getNext())!=null; cur = temp);
        cur.setNext(node);
    }
    size++;
}
```

- ☐ Inserting a node at the beginning of the list
- ☐ Deleting a node at the beginning of the list
- ☒ Inserting a node at the end of the list
- ☐ Deleting a node at the end of the list



✓ What is the time complexity to insert an element to the rear of a LinkedList(head pointer given)?

*1/1

- ☒ $O(n)$
- ☐ $O(1)$
- ☐ $O(\log n)$
- ☐ None



✓ Entries in a stack are "ordered". What is the meaning of this statement? * 1/1

- ☐ A collection of stacks is sortable
- ☐ Stack entries may be compared with the '<' operation
- ☐ The entries are stored in a linked list
- ☒ There is a Sequential entry that is one by one



✓ Which elements not in middle but can be inserted or deleted at/from both *1/1 the ends?

- ☐ Circular queue
- ☐ Priority queue
- ☐ Queue
- ☐ DE queue
- ☒ All of these



This content is neither created nor endorsed by Google. - [Terms of Service](#) - [Privacy Policy](#)

Google Forms



